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Integrating price and textual data for next-day stock movement prediction: a study using StockNet dataset

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Abstract

This research enhances next-day stock movement prediction by integrating price data with social media text from Twitter using the StockNet dataset. We employ diverse preprocessing techniques and modeling approaches, including advanced embedding methods, sentiment analysis, and generative adversarial networks (GANs). Results reveal notable performance trends across embedding techniques and sentiment-based methodologies. Our adversarially fine-tuned BERT model, trained on the Financial Phrasebank dataset, achieves top performance with 74.2% accuracy and 74.1% macro F1, surpassing other techniques. Introducing thresholds significantly improves model performance, with accuracy increasing from 70 to 76.4% and macro F1 score from 70 to 76.3%. Ensemble voting methods further enhance predictive accuracy, reaching 77% with thresholds applied. The integration of GANs provides modest additional improvements. Our approach achieves an F1 score of 78.5%, substantially outperforming previous benchmarks on the same dataset (57.5–60.5%). These findings underscore the importance of data integration, sophisticated modeling methodologies, and careful threshold optimization in achieving accurate and timely stock market predictions.

Keywords Stock movement forecasting, Sentiment analysis, BERT, GANs, Data preprocessing, Predictive modeling, PCA, Classification models, Synthetic data generation

1 Introduction

In today's fast-paced financial landscape, accurately predicting stock market behavior is essential for investors seeking to make informed decisions. The dynamic nature of the stock market, driven by continuous news updates and global economic shifts, makes prediction a challenging yet crucial task. Traditional methods of stock prediction have relied heavily on historical price data and technical indicators. While these methods provided foundational insights, they often fell short in capturing the rapid and complicated influences affecting the market.

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This paper presents a detailed analysis of the effectiveness of various embedding techniques, adversarial training, and modeling approaches for predicting stock movements. It encompasses the performance evaluation of pre-trained embedding models such as BERT [1] and FinBERT [2], alongside Convolutional Neural Networks (CNN) and traditional methods like Continuous Bag of Words (CBOW) and Skip-gram embeddings. The study also examines the impact of threshold-based models on predictive performance and explores the enhancements brought by integrating Generative Adversarial Networks (GANs). Additionally, it delves into the benefits of adversarial training in fine-tuning sentiment analysis models, enhancing their accuracy and robustness.

Moreover, the increasing prevalence of social media has introduced new dimensions to market analysis. Platforms like Twitter (now X) offer real-time insights into public sentiment, often preceding traditional news sources. In this study, Twitter serves as a critical source of textual data, capturing real-time public sentiment and discussions about stocks via the StockNet dataset [3]. Tweets from 2014–2015, aligned with U.S. market hours (e.g., 9 am March 24th to 9 am March 25th for predicting March 25th movements), are processed to extract sentiment scores and embeddings, which are combined with technical indicators to enhance predictive accuracy. This real-time data enables the model to capture market sentiment shifts that traditional news may lag, as demonstrated in prior work [4]. Despite the potential of textual data, its application in stock prediction remains underexplored, particularly in the context of advanced techniques such as GANs and adversarial training. This paper investigates the integration of textual data from social media platforms into stock movement prediction models, utilizing advanced NLP techniques along with CNNs. By exploring adversarial training for sentiment analysis models and GANs for improving prediction accuracy, this approach aims to provide a more robust framework for trend prediction.

The primary objectives of this research are to investigate the effectiveness of different machine learning and deep learning techniques in stock market prediction, develop a novel approach to integrate predictive analytics, and enhance trend prediction models through the integration of textual data and advanced NLP techniques. By addressing the limitations of previous research, such as the exclusion of sentiment analysis-based methods [5], modest model performance and lack of feature engineering [6], real-world implementation challenges and a price-only approach [7], and the absence of defense models [8], this study aims to bridge the gap between traditional financial models and modern data-driven approaches.

Through comprehensive evaluation and comparison, this research seeks to identify the most effective techniques for accurate stock movement prediction, offering valuable insights for investors and financial analysts. By leveraging diverse data sources and advanced techniques, this study aims to empower investors and financial analysts with the tools necessary to navigate the complexities of today's investment landscape with confidence and precision.

2 Literature review

Nguyen et al. [4] explored stock price movement prediction using sentiment analysis from social media, focusing on topic-specific sentiments rather than overall mood. By combining extracted sentiments related to company-specific topics with traditional stock data, their model improved accuracy by 2.07% over one year across 18 stocks, with

a significant 9.83% improvement for harder-to-predict stocks. This approach demonstrated the effectiveness of incorporating topic-sentiment features.

Gunduz et al. [7] forecasted the directional movement of intraday prices for the Borsa Istanbul Index by leveraging historical prices and technical indicators. Their prediction model utilized the preceding 80 h of price data to anticipate the movement of the subsequent price. The model architecture included two parallel CNN layers, each processing the same input data to capture distinct relationships within the sequence. Additionally, a separate CNN layer was employed to merge the outputs from the parallel CNNs.

Gálvez and Gravano [9] explored the predictive potential of online text data from an Argentinian stock message board. By employing machine learning techniques, they demonstrated that integrating message board data improved prediction accuracy, suggesting its complementarity to existing predictive features. Their findings underscored the value of incorporating online text data into stock market prediction systems, providing valuable insights for future research in this area.

Li and Shah [10] investigated the predictive capabilities of investor sentiment indicators on stock market changes by creating a sentiment lexicon and sentiment-oriented word embeddings from StockTwits, a financial social network. This research is the first to develop sentiment-oriented embeddings specifically for the stock market, using neural networks to learn word polarity through sentiment prediction. Experimental results showed that their lexicon outperformed existing state-of-the-art lexicons, and the sentiment-oriented embeddings were more effective than general word embeddings for financial sentiment analysis.

Wang et al. [11] presented a study focused on developing a financial expert system aimed at predicting short-term stock prices by leveraging various online platforms such as Google and Wikipedia. The system integrates historical stock prices, technical indicators, sentiment analysis of financial news articles, Google search trends, and Wikipedia page visit trends. Through a structured approach to data preparation, the authors train four machine learning ensemble methods: neural network regression, support vector regression, boosted regression tree, and random forest regression.

Xu and Cohen [8] introduced StockNet, a deep generative model for stock movement prediction. It utilizes text and price signals, addressing market stochasticity, chaotic information, and temporal dependencies. StockNet achieves state-of-the-art performance by directly learning from data, incorporating predictive dependencies, and using a hybrid objective with temporal auxiliary. Experimental results demonstrate its effectiveness in outperforming strong baselines on a new dataset.

Picasso et al. [12] tackled the stock market prediction problem by combining technical and fundamental analysis using machine learning techniques. Their model predicted market trends for a portfolio of the top 20 NASDAQ100 companies by leveraging both technical indicators and sentiment extracted from news articles. The approach achieved an 85.2% annualized return during a high-frequency trading simulation, highlighting the effectiveness of incorporating sentiment and technical data in stock prediction.

Sawhney et al. [6] introduced MAN-SF, a neural model for stock forecasting that integrates financial data, social media, and inter-stock relationships. Through experiments on S&P 500 data and tweets, they demonstrate its practical applicability for investment decisions. MAN-SF outperforms baseline methods by jointly optimizing learning across multiple data modalities, achieving high predictive power and profitability. The study

highlights the importance of multimodal signals and discusses future research directions, including incorporating additional data sources and improving scalability for cross-market scenarios.

Xie et al. [13] investigated the vulnerability of stock prediction models to adversarial attacks using social media data. They experiment with various attack configurations to deceive three stock prediction models, focusing on adversarial generation with semantics and budget constraints. By formulating the task as a combinatorial optimization problem, they demonstrate consistent success rates and significant monetary losses in trading simulations with perturbed but semantically similar tweets.

Fazlija and Harder [14] focuses on utilizing advanced transformer models, particularly BERT, which are fine-tuned on financial text to forecast the S&P 500's price movement by analyzing news sentiment from sources like Bloomberg and Reuters. These sentiment scores, integrated with time series data, serve as features in a random forest classifier, showing that news-based sentiment is highly effective in predicting stock market trends.

Wu et al. [15] proposes a framework that leverages generative adversarial networks (GANs) combined with the piecewise linear representation (PLR) technique to predict stock trading actions such as buying, selling, and holding. By utilizing GANs, the model addresses the overfitting problem often encountered in traditional supervised learning by generating additional samples that enhance learning. Experimental results reveal that this GAN-based framework outperforms long short-term memory (LSTM) models in stock trading action prediction.

Gangopadhyay and Majumder [5] investigated how financial news articles, particularly those published during market hours, influence stock market movements. The proposed parallel CNN model effectively combines text and price data, achieving high prediction accuracy. Results indicate that incorporating news article text significantly enhances predictions, underscoring the importance of context. The study offers insights into leveraging contextual information for market prediction and other classification tasks.

3 Methodology

The methodology outlined herein delineates a comprehensive approach to forecasting stock movements, amalgamating numerical and textual data through preprocessing and modelling techniques. The foundation of this methodology rests upon adept data handling, where a robust dataset is curated to facilitate predictive modelling. This encompasses an intricate process of calculating technical indicators from price data, encapsulating underlying market trends and patterns. For instance, in forecasting for a specific date like March 25th, technical indicators derived from the preceding day's price data (March 24th) are combined with tweets spanning from 9 am on March 24th to 9 am on March 25th, synchronising with the opening time of the US stock market. These resulting features, comprising numerical indicators and textual data, serve as inputs for predicting the directional movement of stocks.

3.1 Classification approaches

In this section, two distinct approaches are delineated for predicting stock direction. The initial approach categorises movements as positive if they are greater than or equal to 0% change and negative otherwise. In contrast, the second approach employs a threshold-based classification, again a binary classification problem, where movements above 0.55

are categorised as positive and those below -0.5 as negative. By focusing only on significant stock movements, minor fluctuations are excluded, improving the signal-to-noise ratio. This thresholding method, initially utilised in [8], is substantiated by its practical implications, particularly regarding trading strategies. By focusing on substantial differences between closing prices on consecutive days, this approach facilitates discernible trading decisions, accounting for tax implications and trade charges.

3.2 Preprocessing techniques

The preprocessing stage is pivotal in refining textual data to extract meaningful insights. Techniques such as lemmatisation, stop words removal, and the elimination of unwanted characters are applied to streamline the textual column. Concurrently, handling technical indicators is optimised to ensure they are range-bound and independent of specific stocks or prices. This ensures that predictive models capture patterns inherent within the technical indicators and textual data without being unduly influenced by individual stock characteristics. The fusion of these processed data streams, encompassing technical indicators and textual features from tweets, into a unified dataframe comprising 87 stocks form the bedrock for subsequent modelling and analysis.

3.2.1 Approaches for preprocessing of technical indicators

We calculate the following Technical Indicators, and temporal Harmonics for each selected asset: Simple Moving Averages, Exponential Moving Averages, Relative Strength Index, Moving Average Convergence Divergence, Bollinger Bands, Average Directional Index, Average True Range, Chaikin A/D Oscillator, Hilbert Transform Dominant Cycle Period, Hilbert Transform Dominant Cycle Phase, Sine and Cosine values of day of the week, date, and month.

We conducted feature engineering on historical asset prices. First, we standardised technical indicators like moving averages (e.g., SMA-21, EMA-12) by calculating their percentage differences from the 'Adj Close' price, enabling more effective trend analysis and deviation tracking. Additionally, we transformed the Bollinger Bands by replacing the original values of the Upper, Middle, and Lower Bands with their respective percentage differences from the corresponding Adjusted Close price. Furthermore, we computed the MACD (Moving Average Convergence Divergence) indicator and used its histogram values as standardised features obtained by subtracting the MACD line from the signal line. This rigorous feature engineering process ensures that all technical indicators are on a consistent scale, facilitating more accurate trend detection in our research analysis.

3.2.2 Approaches for preprocessing of textual data

In parallel with handling numerical data, the methodology extends to the intricate task of processing textual data through two distinct approaches.

1. *Sentiment analysis*: The first approach integrates sentiment analysis, leveraging state-of-the-art techniques to discern the sentiment conveyed within tweets. Using the Financial Phrase Bank dataset , a BERT model is employed, fine-tuned on financial discourse. This fine-tuning process encompasses adversarial techniques, augmenting the dataset with synonyms, semantic variations, and noise, including typos and word order permutations, as shown in Algorithm 1. The resultant model, adept at classifying

sentiment into positive, neutral, and negative categories, is then applied to the textual column of the dataset generated in prior steps. These sentiment predictions replace the original textual data, serving as enriched features for subsequent analysis. Along with that, we even tried sentiment analysis using TextBlob, a well-known sentiment analysis library and compared that with our sentiment analysis model, as shown in Fig. 1.

2. *Contextual embeddings*:
The second approach revolves around generating contextual embeddings from textual data, harnessing various techniques. Pre-trained models such as BERT and FinBERT [16], along with traditional methods like TF-IDF, are employed to derive embeddings capturing the nuanced context of each sentence. For BERT and FinBERT, sentence-level embeddings are extracted using the [CLS] token, representing each tweet as a fixed-length vector (e.g., 768 dimensions for BERT). These embeddings are fed into a 1D Convolutional Neural Network (CNN) with 64 filters, a kernel size of 3, ReLU activation, and max-pooling to capture local patterns and interactions across tweet sequences for a given stock, followed by a dense layer for classification. This CNN layer enhances feature extraction by modeling temporal or structural relationships

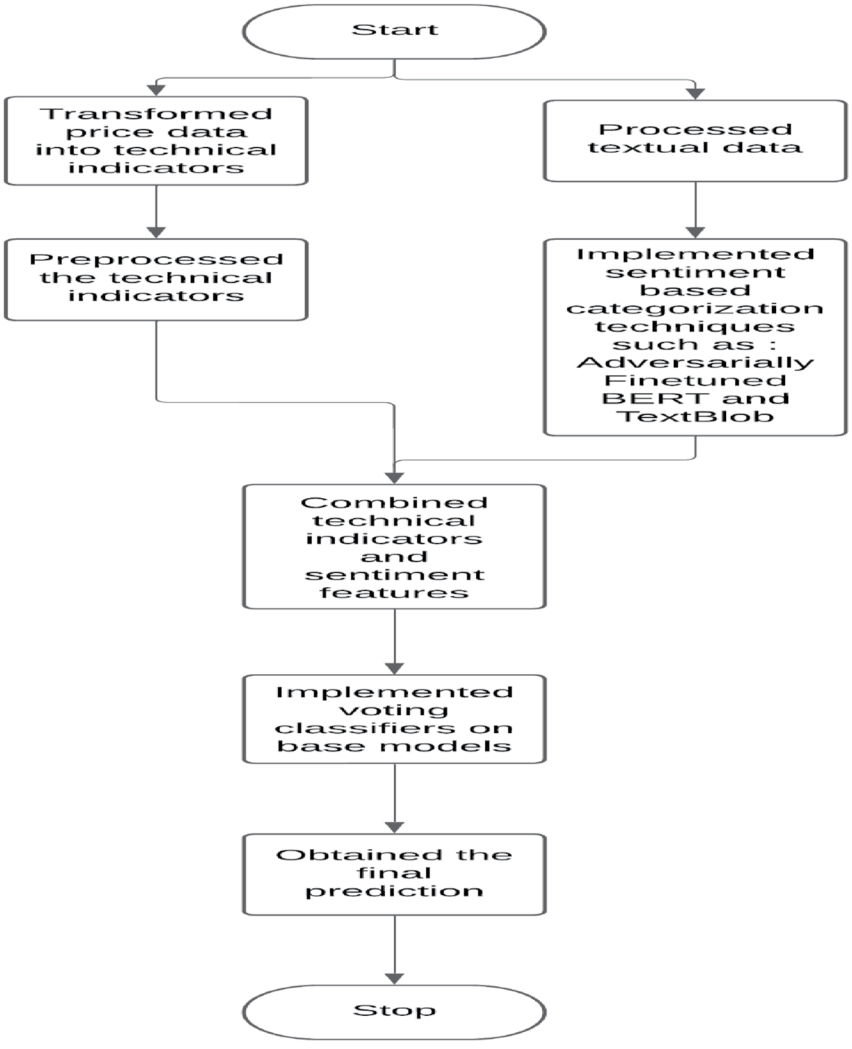


Fig. 1 Sentiment analysis based approaches

in the sequence of tweet embeddings, as seen in similar financial NLP applications [6]. For TF-IDF, the CNN operates on the sequence of TF-IDF vectors, which is more conventional. Additionally, word2vec [17] approaches such as SkipGram and CBOW are utilized, omitting the need for CNNs as they inherently consider context during embedding generation. These embeddings, ranging from high-dimensional representations, are further distilled into aggregated sentence vectors by averaging over each row, resulting in a single vector per tweet. To manage computational complexity and extract salient features, Principal Component Analysis (PCA) is applied to reduce the dimensionality of these vectors to a concise length of four. The technical indicators, alongside the PCA components, collectively serve as independent features, forming a comprehensive feature set for predictive modeling as shown in Fig. 2.

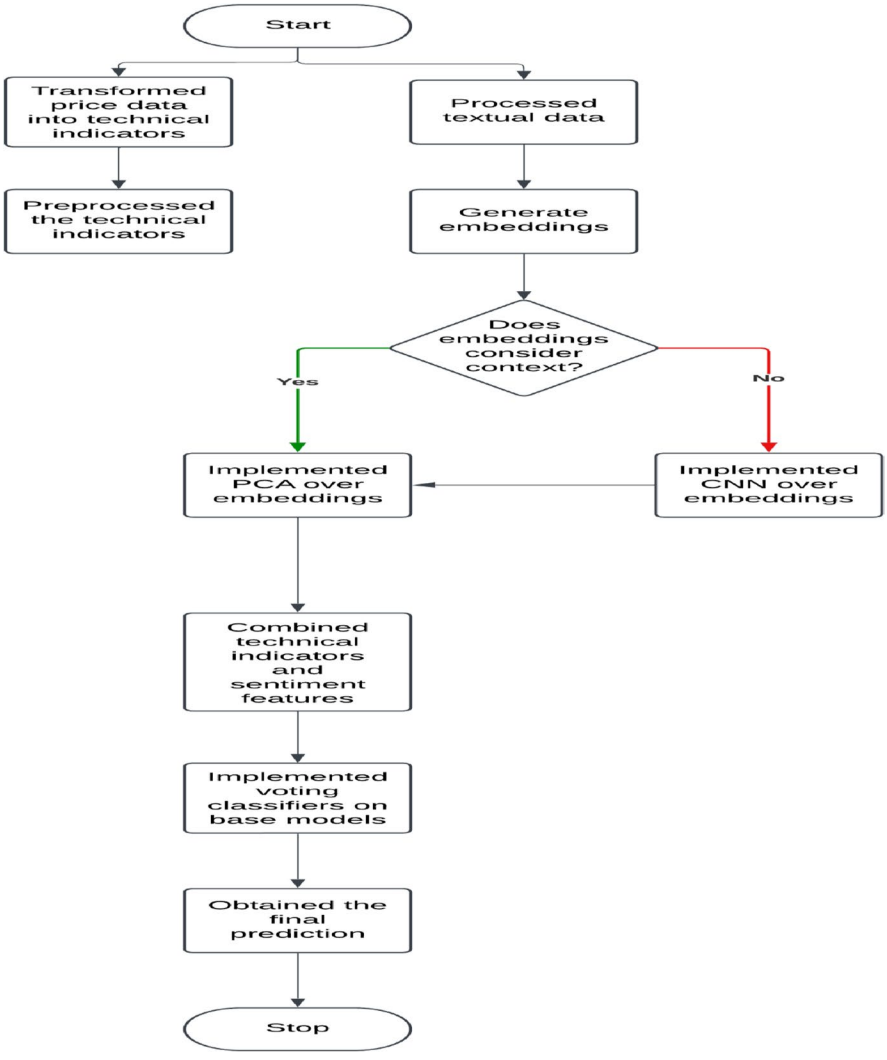


Fig. 2 Embeddings based approaches

Read and process input files:
  Read each file and split data into lines.
  Extract text and sentiment information from each line.
  Convert sentiment classes to numeric values.
  Clean text data by removing special characters and extra spaces.
Split Data:
  Split the processed data into training and testing sets.
Synonym Augmentation:
  For each text in the training data:
    Tokenise the text into words.
    For each word:
      Find synonyms using WordNet.
      Replace the word with a random synonym.
Introduce Typos:
  Apply to 20% of the rows in the text data:
    Introduce typos in 30% of the words within each selected text.
Shuffle Words:
  Apply to 20% of the rows in the textual data:
    Shuffle the words within each selected text randomly using Python's `shuffle()` function.
Mask Words:
  Mask words in each text to generate masked versions.
Train and Evaluate Model:
  Train a BERT-based sequence classification model on the augmented training data.
  Evaluate the model on the testing set using classification metrics.

Algorithm 1 Data augmentation and adversarial fine-tuning of BERT architecture

3.3 Generative adversarial networks (GANs)

In tandem with the methodologies mentioned above, we introduce Generative Adversarial Networks (GANs) [18] to enhance the predictive capabilities of our model. GANs are employed to address the limited textual data available in the StockNet dataset (covering 2014–2015), which was insufficient for robust machine learning model training. The GAN generates synthetic feature vectors that combine preprocessed technical indicators (e.g., standardized SMA, RSI, MACD) and sentiment scores derived from the adversarially fine-tuned BERT model applied to tweets. These synthetic vectors, with the same dimensionality as the real data (e.g., 20 technical indicators + 3 sentiment scores), are integrated into the training dataset to improve model robustness and generalization by exposing it to a wider range of scenarios, reducing overfitting.

GANs have been successfully applied in time-series and sequence modeling tasks, such as financial simulation [15] and trajectory forecasting [19]. Their ability to model complex data distributions makes them suitable for augmenting financial datasets. However, GAN training can face challenges like mode collapse and instability. To mitigate these, we used batch normalization and skip connections in both the Generator and Discriminator, with a learning rate of 0.0002 and 1000 epochs, monitoring discriminator loss to ensure convergence.

1. *Generator*: The Generator accepts a 100-dimensional noise vector as input, reshaped to fit convolutional layers. These undergo a series of transformations, including convolutional operations with ReLU activation, max-pooling to reduce spatial

dimensions, and skip connections to facilitate information flow. Batch normalization is applied to stabilize training. The Generator outputs synthetic feature vectors with the same dimensionality as the input actual data, achieved through a dense layer with a tanh activation function.

2. *Discriminator*: The Discriminator operates on input data of the same shape as the Generator's output. It starts with a dense layer featuring LeakyReLU activation, followed by batch normalization and skip connections to optimize information flow. Subsequent layers continue this pattern, concluding with a dense layer utilizing sigmoid activation for binary classification, discerning whether the input data is real or synthetic. Through this adversarial training process, the Generator and Discriminator iteratively improve, resulting in synthetic data that closely resembles real data distributions.

The GAN-augmented dataset, combining real and synthetic feature vectors, is used to retrain classifiers (e.g., Random Forest, LightGBM), enhancing their performance, as shown in Tables 4 and 5. The modest improvements (e.g., +0.01 in soft-voting accuracy with thresholds) justify the added complexity in the context of stock prediction, where small gains can translate to significant financial outcomes.

3.4 Model optimisation and evaluation

Moving forward, the methodology extends to implementing grid search cross-validation (GridSearchCV) on a myriad of classifiers, including Naive Bayes (NB), Decision Trees (DT), Random Forest (RF), K-Nearest Neighbors (KNN), Logistic Regression (LR), LightGBM (LGBM) [20], Gradient Boosting (GB) [21], Catboost (CB) [22], Linear Discriminant Analysis (LDA), and Quadratic Discriminant Analysis (QDA).

For Gradient Boosting, the parameters used are 'n_estimators', 'learning_rate', and 'max_depth'. In the case of Decision Trees, the parameters include 'max_depth' and 'min_samples_split'. When tuning LightGBM, the relevant parameters are 'n_estimators', 'learning_rate', and 'max_depth'. For CatBoost, the parameters considered are 'iterations', 'learning_rate', and 'depth'.

For Support Vector Machines, the grid search focuses on 'C', 'kernel', and 'gamma'. Logistic Regression is tuned using 'C' and 'penalty'. Finally, for Random Forest, the parameters include 'n_estimators', 'max_depth', and 'min_samples_split'.

The data is split into training and testing sets with a ratio of 80–20%. The best performing classifiers, typically the top three or four models, are selected for further analysis. These models are then employed in hard-voting and soft-voting classifiers to bolster robustness and mitigate individual model biases. Subsequently, classification reports are generated to compare the performance of all classifiers, including the voting classifiers and Artificial Neural Networks. This comprehensive evaluation aids in identifying the most effective models for accurately predicting stock movements, offering valuable insights for investors and financial analysts.

3.5 Real-world implementation and scalability

The proposed models demonstrate strong performance and scalability, making them suitable for real-world stock market prediction scenarios. For example, a positive sentiment score derived from tweets like "$XYZ new product launch is a game-changer!" may lead to a "buy" prediction considering technical indicators forming a bullish patterns as

well, while negative sentiment from tweets like "$XYZ facing regulatory issues" could trigger a "sell" prediction, guiding trading decisions in real-world applications. However, for more trustworthy results, especially in practical applications, it is important to include fundamental data such as company financials and macroeconomic indicators, which were unavailable in this study. Integrating this data would enhance the model's ability to capture stock movements more accurately. Experimenting on other datasets is not mandatory because we are implementing machine learning models on technical indicators, not directly on price data. Technical indicators, such as RSI, remain constant across all markets, as a value of 70 indicates overbought and 30 indicates oversold, regardless of the market. Additionally, the models are designed to handle larger datasets without performance degradation, ensuring scalability. Given their structure and computational efficiency, they should scale well for real-time or high-frequency trading scenarios with appropriate infrastructure.

3.6 Hardware requirements

For this study, the models were trained and evaluated using a GPU P100 with 16 GB memory, along with a CPU configuration of 32 GB RAM in a virtual environment. This setup was adequate for handling the computational demands of the various machine learning models and NLP techniques used, including adversarial training and GAN-based approaches.

4 Results

This section provides a comprehensive overview of our investigation into the effectiveness of various embedding techniques and modelling approaches for predicting stock movements. We delve into the performance of pre-trained embedding models such as BERT and FinBERT, in conjunction with Convolutional Neural Networks (CNN), as well as TF-IDF, in conjunction with CNN, to acquire contextual embeddings. Furthermore, we explore the traditional methods of Continuous Bag of Words (CBOW) and Skip-gram embeddings to capture intricate contextual patterns within financial text data. Additionally, we analyse the influence of threshold-based and threshold-agnostic models on predictive performance, aiming to discern their applicability across different prediction tasks.

4.1 Evaluation of embedding techniques and impact of thresholds

In the evaluation of embedding techniques for predicting stock movements, we observed distinct performance trends.

Among the embedding techniques, FinBERT Embeddings combined with CNN yielded the highest average accuracy of 0.73 and an average macro F1 score of 0.7275. This was closely followed by TF-IDF+CNN, which achieved an average accuracy of 0.728 and an average macro F1 score of 0.727. BERT Embeddings+CNN and CBOW embeddings demonstrated competitive performance, with average accuracies of 0.7 and 0.727, respectively. In contrast, Skip-gram embeddings exhibited slightly better performance, with an average accuracy of 0.728 and an average macro F1 score of 0.727. Refer to Tables 1 and 3 for further details.

Table 1 Comparison of various models for stock movement prediction using technical indicators and embedding techniques

Technique	Threshold	Metric	Models								
			ANN	KNN	DT	LGBM	CB	LR	NB	LDA	QDA
FinBERT embeddings + CNN	No	Acc	0.51	0.67	0.66	0.72	0.71	0.6	0.53	0.59	0.51
		F1	0.34	0.665	0.65	0.715	0.71	0.58	0.46	0.54	0.37
	Yes	Acc	0.5	0.72	0.72	0.79	0.77	0.64	0.58	0.64	0.53
		F1	0.375	0.72	0.72	0.785	0.77	0.64	0.57	0.64	0.35
BERT embeddings + CNN	No	Acc	0.49	0.67	0.66	0.72	0.71	0.6	0.52	0.59	0.53
		F1	0.33	0.665	0.66	0.715	0.71	0.595	0.47	0.59	0.53
	Yes	Acc	0.5	0.72	0.72	0.79	0.77	0.64	0.59	0.64	0.5
		F1	0.335	0.72	0.72	0.785	0.77	0.64	0.54	0.635	0.35
TF-IDF + CNN	No	Acc	0.49	0.64	0.66	0.72	0.71	0.6	0.53	0.59	0.57
		F1	0.37	0.635	0.66	0.715	0.71	0.595	0.47	0.59	0.56
	Yes	Acc	0.5	0.68	0.72	0.79	0.77	0.64	0.59	0.64	0.52
		F1	0.34	0.685	0.715	0.79	0.77	0.635	0.53	0.635	0.45
CBOW	No	Acc	0.63	0.66	0.66	0.72	0.7	0.6	0.55	0.59	0.55
		F1	0.625	0.665	0.655	0.715	0.7	0.595	0.485	0.59	0.485
	Yes	Acc	0.68	0.71	0.72	0.79	0.77	0.64	0.59	0.64	0.57
		F1	0.665	0.71	0.72	0.785	0.765	0.64	0.53	0.635	0.56
Skip-gram	No	Acc	0.62	0.66	0.66	0.72	0.71	0.6	0.52	0.59	0.55
		F1	0.62	0.66	0.66	0.715	0.71	0.595	0.49	0.59	0.455
	Yes	Acc	0.67	0.72	0.72	0.79	0.77	0.64	0.54	0.63	0.5
		F1	0.67	0.715	0.72	0.785	0.77	0.635	0.535	0.635	0.37

Table 2 Comparison of various models for stock movement prediction using technical indicators and sentiment analysis techniques

Technique	Threshold	Metric	Models						
			ANN	GB	DT	LGBM	CB	LR	RF
Adversarially finetuned BERT	No	Acc	0.67	0.7	0.66	0.7	0.71	0.59	0.71
		F1	0.65	0.695	0.64	0.7	0.71	0.56	0.705
	Yes	Acc	0.74	0.76	0.72	0.77	0.78	0.64	0.78
		F1	0.74	0.76	0.715	0.77	0.78	0.635	0.78
TextBlob	No	Acc	0.71	0.69	0.67	0.7	0.7	0.6	0.7
		F1	0.7	0.685	0.67	0.695	0.7	0.59	0.7
	Yes	Acc	0.71	0.76	0.73	0.77	0.77	0.64	0.78
		F1	0.705	0.76	0.725	0.77	0.77	0.635	0.775

4.2 Evaluation of sentiment-based approaches and impact of thresholds

We now focus on sentiment-based approaches for stock movement prediction.

Among the sentiment-based approaches, Adversarially Fine-tuned BERT emerged as the top-performing technique, with an average accuracy of 0.742 and an average macro F1 score of 0.741. TextBlob followed closely behind, achieving an average accuracy of 0.738 and an average macro F1 score of 0.736. Refer to Tables 2 and 3 for further details.

4.3 Comparison and best techniques

Comparing the best-performing techniques from embedding and sentiment categories, Adversarially Fine-tuned BERT outperformed FinBERT Embeddings+CNN with a higher average accuracy (0.742 vs. 0.73) and macro F1 score (0.741 vs. 0.7275). It also showed superior performance in both hard and soft-voting. The superior performance

Table 3 Performance metrics of various embedding techniques with different ML models and ensemble methods

Embedding technique (threshold)	ML models				Hard voting		Soft voting	
	Model 1	Model 2	Model 3	Model 4	Accuracy	F1 score	Accuracy	F1 score
Fin-BERT embeddings + CNN (no)	KNN	LGBM	CB	-	0.72	0.715	0.71	0.715
Fin-BERT embeddings + CNN (yes)	KNN	DT	LGBM	CB	0.78	0.775	0.78	0.775
BERT embeddings + CNN (no)	KNN	DT	LGBM	CB	0.7	0.7	0.69	0.69
BERT embeddings + CNN (yes)	KNN	DT	LGBM	CB	0.77	0.77	0.76	0.76
TF-IDF + CNN (no)	DT	LGBM	CB	–	0.71	0.715	0.69	0.69
TF-IDF + CNN (yes)	DT	LGBM	CB	–	0.78	0.785	0.76	0.765
CBOW (no)	KNN	DT	LGBM	CB	0.71	0.705	0.7	0.695
CBOW (yes)	KNN	DT	LGBM	CB	0.76	0.755	0.76	0.76
Skip-gram (no)	KNN	DT	LGBM	CB	0.7	0.695	0.68	0.685
Skip-gram (yes)	KNN	DT	LGBM	CB	0.76	0.75	0.76	0.76
Adversarially finetuned BERT (no)	GB	LGBM	CB	RF	0.71	0.715	0.72	0.715
Adversarially finetuned BERT (yes)	GB	LGBM	CB	RF	0.78	0.785	0.78	0.78
TextBlob (no)	GB	LGBM	CB	RF	0.7	0.715	0.71	0.71
TextBlob (yes)	GB	LGBM	CB	RF	0.77	0.77	0.77	0.775

of Adversarially Fine-tuned BERT is attributed to the adversarial fine-tuning process, which augments the Financial PhraseBank dataset with synonyms, typos, and word order permutations (Algorithm 1). This enhances robustness to the noisy and varied nature of Twitter data, which FinBERT, despite its financial pre-training, may not fully capture without such targeted fine-tuning. Based on these metrics, Adversarially Fine-tuned BERT is the best-performing technique.

4.4 Impact of thresholds on predictive performance

The impact of thresholds on predictive performance is critical in evaluating various embedding and sentiment techniques for stock movement prediction. When assessing predictive performance without thresholds, the average accuracy across the top three classifiers of different embedding and sentiment techniques stands at 0.7, reflecting the overall predictive capability of these models. Similarly, the macro F1 score, which considers both precision and recall across multiple classes, also averages 0.7, indicating a balanced performance in terms of classification accuracy.

Furthermore, when evaluating the predictive performance without thresholds, the average of hard-voting and soft-voting across all different embedding and sentiment techniques is 0.7. This suggests that the ensemble methods employed, incorporating the collective decisions of multiple classifiers, exhibit consistent accuracy across various techniques.

In contrast, with the introduction of thresholds, there is a notable improvement in predictive performance. The average accuracy across the top 3 classifiers of different embedding and sentiment techniques increases to 0.764, indicating a refined ability to classify stock movements correctly. Similarly, the macro F1 score improves to 0.763,

Table 4 Impact of GAN on predictions for the data with thresholds

	Hard voting	Soft voting
Accuracy	0.78	0.79 (+0.01)
Macro F1	0.78 (−0.005)	0.785 (+0.005)

Table 5 Impact of GAN on predictions for the data without thresholds

	Hard voting	Soft voting
Accuracy	0.71	0.72
Macro F1	0.715	0.72 (+0.005)

signifying enhanced precision and recall in predicting stock movements across multiple classes.

Moreover, the average of hard-voting and soft-voting across all different embedding and sentiment techniques demonstrate a further increase to 0.77 when thresholds are applied. This underscores the effectiveness of threshold-based models in enhancing the consensus among individual classifiers and improving the overall predictive accuracy.

4.5 Impact of GAN implementation on results

Upon integrating GAN with the Adversarially Fine-tuned BERT model, which previously exhibited optimal performance, marginal enhancements were observed in the results. The GAN generates synthetic feature vectors combining technical indicators and BERT-derived sentiment scores, augmenting the training dataset. The added computational complexity of GAN training, requiring a GPU P100 with 16GB memory, is justified by the marginal improvements (e.g., +0.01 in soft-voting accuracy with thresholds, as shown in Table 4), as small gains in stock prediction can lead to significant financial outcomes. Simpler augmentation techniques, such as synonym replacement, were less effective, supporting the use of GANs. Table 4 displays the accuracy and F1 scores of hard-voting and soft-voting classifiers, utilizing thresholds and the variations post-GAN implementation. Table 5 presents the same metrics without thresholds, alongside changes after GAN incorporation.

4.6 Performance evaluation: benchmarking our results against comparable studies

In comparison with previous research efforts utilising the same dataset, our proposed approach outperforms existing methodologies. Xu and Cohen [8] achieved an F1 score of 57.5% using StockNet, Feng et al. [23] obtained 57% with Adversarial LSTM, and Sawhney et al. [6] reported 60.5% using MAN-SF. Zhao et al. [24] employed a multi-layer features ablation approach on the BERT model to predict stock market trends using expert comments, leveraging a different dataset. Their method addressed BERT's limitations by using sliding windows for long text processing and extracting multi-layer features, achieving an accuracy of 70.3% for the Shanghai Composite Index and 67.6% for the Shenzhen Component Index. Despite the dataset differences, both studies highlight the effectiveness of textual analysis in financial forecasting, with notable improvements in prediction accuracy. Zhang and Wang [25] introduced a stock trend prediction method combining wavelet transform and a Temporal Convolutional Network (TCN), alongside sentiment analysis using a GRU network for stock news data. They utilized a dataset sourced from expert stock comments, which were manually annotated to classify

sentiments as 'positive' or 'negative.' Their approach resulted in an accuracy of 60.3% for Gree Electric, 59.62% for ZTE, and 62.15% for AAPL.

Our approach, which integrates adversarially fine-tuned BERT, GAN-augmented data, and threshold-based classification, achieves an F1 score of 78.5% and accuracy of 78% on the StockNet dataset, significantly surpassing these benchmarks. The superior performance is attributed to the robustness of our adversarial fine-tuning and the augmentation of limited data via GANs, though limitations include the absence of fundamental data (e.g., company financials).

4.7 Case study: sentiment impact on predictions

To illustrate the impact of sentiment analysis, we present a case study using example tweets. For stock $XYZ on March 24, 2015, the tweet "$XYZ new product launch is a hit!" was classified as positive (sentiment score: 0.85) by the adversarially fine-tuned BERT model, contributing to a "buy" prediction. Conversely, for stock $ABC, the tweet "$ABC facing supply chain issues" was classified as negative (sentiment score: −0.78), leading to a "sell" prediction, along with the technical indicators data. These examples demonstrate how sentiment scores, combined with technical indicators, drive accurate stock movement predictions, as reflected in the high F1 score (78.5%) reported in Table 2.

5 Conclusion

This study presents a comprehensive investigation into the integration of price data and textual information from social media, specifically Twitter, for next-day stock movement prediction. Leveraging the StockNet dataset [3], which combines stock prices and associated tweets, we employed various preprocessing techniques and modelling approaches to develop robust predictive models. Our findings underscore the significance of combining numerical and textual data sources to enhance the accuracy and timeliness of stock movement predictions.

Through empirical experimentation, we evaluated various embedding techniques, including BERT, FinBERT, SkipGram, CBOW, and TF-IDF, in conjunction with different classifiers. We observed distinct performance trends among embedding techniques and sentiment-based approaches, with Adversarially Fine-tuned BERT emerging as the top-performing technique. Integrating adversarial training techniques proved instrumental in fine-tuning sentiment analysis models, enhancing their accuracy and robustness. While the Adversarially Fine-tuned BERT model remains the top performer, adding Generative Adversarial Networks (GANs) showed some minor improvements. This suggests that GANs could offer more refinements for future predictive analysis.

Furthermore, our study explored the impact of thresholds on predictive performance, highlighting the effectiveness of threshold-based models in improving classification accuracy. By introducing thresholds, we observed notable enhancements in accuracy, F1 score, and ensemble voting across different embedding and sentiment techniques. This underscores the importance of carefully selecting and optimizing threshold parameters to maximise predictive capabilities.

In conclusion, our research demonstrates the efficacy of integrating price and textual data for next-day stock movement prediction. By leveraging advanced NLP techniques, adversarial training, and ensemble learning, we have developed models capable of

capturing nuanced market sentiments and trends. These findings offer valuable insights for investors and financial analysts, emphasizing the importance of leveraging real-time data sources and sophisticated modeling approaches for accurate and timely stock market predictions.

Author contributions
D.M. worked on the technical aspects of the study and contributed to the literature review. A.J. contributed to the literature review and provided supervision throughout the project. C.J. provided domain-specific inputs in the area of finance and conducted research on current methods in the field.

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Data availability
The datasets used in this study are publicly available. The Financial PhraseBank dataset can be accessed at https://huggingface.co/datasets/takala/financial_phrasebank, and the StockNet dataset is available at https://github.com/yumoxu/stocknet-dataset. The StockNet dataset, sourced from Twitter (now X), is anonymized and was collected in accordance with Twitter's API terms and conditions (2014–2015), ensuring no personally identifiable information is included. Our use complies with ethical research standards and journal guidelines on copyright and privacy. While the original datasets, Financial PhraseBank and StockNet, are publicly available, the adversarially generated and preprocessed dataset used in our experiments was derived by augmenting and transforming the original data using the methodology described in the manuscript. This final dataset is not publicly hosted due to its customized nature but is available from the corresponding author upon reasonable request, in accordance with reproducibility best practices.

Declarations

Ethics approval and consent to participate
Not applicable

Consent to publcation
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The authors declare no competing interests

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